Project 1: Income Classification with Naive Bayes

Machine Learning

Question 1: Supervised Model Training

1.1 Prior Probabilities

Class	P(c)
<=50K	0.754
>50K	0.246

The training set has a significant class imbalance of roughly 3:1. This could have several effects on the behaviour of the classifier:

- The model starts with a strong bias towards the negative class <=50K. A person might need strong evidence in their features to shift the prediction towards >50K.
- Likely high recall for the class <=50K but lower recall for the class >50K (missing some high earners by defaulting to the majority class)
- Unsure or borderline cases will tend to be classified as <=50K most of the time.

1.2 Continuous Feature Parameters

Feature	Mean (<=50K)	Std (<=50K)	Mean (>50K)	Std (>50K)
age	36.99	13.72	43.96	10.37
education-num	9.64	2.45	11.6	2.37
capital-gain	151.49	968.34	3542.33	13394.02
capital-loss	56.86	319.34	211.16	616.33
hours-per-week	39.45	11.9	45.66	10.42

Education-Num shows the greatest separation between two classes based on its mean and standard deviation (compared using mean / average std). This can apply in real-life since a higher education correlates well with higher-paying jobs. Age is also a great separator since income tends to grow with career experience.

Capital features are highly skewed and very noisy individual predictors despite having a big mean difference.

1.3 Most Predictive Categorical Values

Top 5 most predictive values for >50K:

Rank	Feature	Value	R
1	education	Prof-school	8.06
2	education	Doctorate	6.95

3	education	Masters	3.48
4	native-country	Taiwan	3.47
5	workclass	Self-emp-inc	3.25

Top 5 most predictive values for <=50K:

Rank	Feature	Value	R
1	relationship	Own-child	0.05
2	occupation	Priv-house-serv	0.10
3	education	9th	0.12
4	occupation	Other-service	0.14
5	relationship	Other-relative	0.15

For >50K, yes they seem reasonable. Professional educational degrees such as masters, doctorates, and people in those professions, such as doctors and lawyers, are generally associated with a high income. Self-employed people indicate to be successful business owners. There are a few Taiwanese high-earning individuals in the data, so that might inflate the R ratio.

For <=50K, that seems reasonable as well. Professions such as household service are low-wage and education only to 9th grade might limit career opportunities. Own child might mean someone young and still learning or dependent and living with their parents.

Question 2: Supervised Model Evaluation

2.1 Overall Accuracy and Class-Level Metrics

Model: MixedBayesClassifier trained on 100% training data
Overall Accuracy = 0.830

Confusion Matrix:

	Predicted >50K	**Predicted <=50K**
Actual >50K	1928	1779
Actual <=50K	801	10568

Classification Metrics:

Class	**Precision**	**Recall**	**F1-Score**
<=50K *(-ve class)*	0.856	0.926	0.891
>50K *(+ve class)*	0.706	0.520	0.599

2.2 Out-of-Vocabulary / Unseen Categories

Only 1 out of 15087 test instances contained an unseen category. It was a native-country value that did not appear in the training data. The implementation handled it through the OrdinalEncoder, which is fitted on the training set and reused for encoding both the test and unlabelled sets. When the encoder sees a category it hasn't seen before, it assigns it a value of -1.

However since CategoricalNB requires non-negative integers, the encode_categorical() function detects these -1 entries and replaces them with 0, mapping the unknown category to an existing one. Because Laplace smoothing (alpha=1.0) is applied, no single category dominates the probability distribution, so the unseen value receives a small probability rather than causing a crash or being ignored. No test instances were skipped, all 15087 were classified successfully.

2.3 High and Low Confidence Predictions

(a) Instances classified as >50K with high confidence:

 # Row indices refer to the preprocessed test set after dropping missing values and resetting the index.

Row	**Age**	**Workclass**	**Education**	**Occupation**	**Hours-per-week**	**…**	**log_R**
6669	42	Self-emp-not-inc	Masters	Sales	80	…	4786.952
7800	29	Self-emp-not-inc	Prof-school	Prof-speciality	60	…	4784.148
7166	44	Self-emp-inc	Bachelors	Exec-managerial	45	…	4788.049
7354	32	Private	Bachelors	Sales	50	…	4785.219
6064	33	Private	Some-college	Adm-clerical	30	…	4773.383

Reporting log_R values since raw R was extremely high (inf).

(b) Instances classified as <=50K with high confidence:

Row	Age	Workclass	Education	Occupation	Hours-per-week	…	log_R
12269	18	Private	10th	Other-service	3	…	-20.802
2993	17	Private	10th	Other-service	12	…	-20.651
9824	17	Private	11th	Other-service	5	…	-20.621
8197	23	Private	Preschool	Other-service	15	…	-20.389
11286	18	Private	11th	Other-service	6	…	-20.344

Reporting log_R values since raw R was extremely low (9.242193e-10).

(c) Instances near the decision boundary (R ~ 1):

Row	Age	Workclass	Education	Occupation	Hours-per-week	…	R
7567	29	Private	Bachelors	Exec-managerial	45	…	1.001
8380	33	Private	Bachelors	Prof-speciality	40	…	0.997
6534	33	Private	Bachelors	Prof-speciality	40	…	0.997
12547	33	Private	Bachelors	Prof-speciality	40	…	0.997
4907	33	Private	Bachelors	Prof-speciality	40	…	0.997

2.4 Discussion

Both high and low confidence predictions seem reasonable. <=50K is easier to classify because low income earners tend to look similar with attributes such as less educated, young and fewer hours per week. High income earners can be trickier because they come in more varieties, for example a self-employed business owner, a young doctor or an old executive. Furthermore, the model starts with a 3:1 bias towards the <=50K class, so it needs strong evidence to classify an instance as >50K.

Question 3: Semi-Supervised Extension (Option 2: Active Learning)

3.1 Instances Selection

Random Instances- Selecting 200 random instances from adult_unlabelled.csv file

Instances which provide the most information- Chosen based on Q1 model's confidence, where the model is uncertain (R is close to 1)

 # Sample of 200 instances selected via uncertainty sampling (R closest to 1).

Instance	Age	Workclass	Education	…	Capital Loss	Hours Per Week
11433	30	Private	Bachelors	…	0	43
11127	33	Private	Bachelors	…	0	40
3257	33	Private	Bachelors	…	0	40
8372	37	Self-emp-inc	Assoc-voc	…	0	60
…	…	…	…	…	…	…
11427	40	Private	Bachelors	…	0	40
7151	22	Private	Assoc-voc	…	1762	40
14978	62	Private	Bachelors	…	0	40
3251	44	Private	Masters	…	0	50

3.2 Implementation Details

Active learning with uncertainty sampling was chosen as the semi-supervised approach. The Q1 model is run on all unlabelled instances to compute posterior ratios and the 200 instances where R is closest to 1 are selected. These sit right on the decision boundary where the model cannot confidently distinguish between the two classes. The reasoning behind this is that labelling these borderline cases gives the model the most useful new information, since it already handles clear-cut cases well. Once selected, the true labels for these 200 instances are revealed from the ground-truth file and they are added to the training set. A new MixedNaiveBayes model is then fitted on this expanded dataset. Random selection was also implemented as a baseline to verify that targeted selection actually outperforms regular Naïve Bayes sampling.

3.3 Hyperparameter Tuning and Validation

A validation set (20% of the supervised training data) was used for all tuning decisions to avoid any leakage from the test set. Three models were compared: the supervised baseline, random selection of 200 instances and uncertainty sampling of 200 instances. Uncertainty sampling was selected as the final strategy based on this result. A default Laplace smoothing parameter of alpha=1.0 was used, which provided stable probability estimates across all category values without over-fitting.

Model	N (50)	N (100)	N (200)	N (300)	N(500)
Random	0.8316	0.8316	0.8322	0.8319	0.8322
Uncertainty	0.8326	0.8326	0.8326	0.8329	0.8329

Model	Overall Accuracy
Supervised (train-validation split)	0.8322
Random	0.8322
Uncertainty	0.8326

3.4 Discussion

The improvement from active learning was very marginal. The validation accuracy moved from 0.832 to 0.833. This was secondary given that 200 additional instances represent roughly 1.5% of the existing training set of around 12000 samples. At that scale, even perfectly chosen instances cannot dramatically shift the learned parameters. This also suggests the supervised model had already learned the key patterns from the data, and the errors it still makes are likely due to the naive independence assumption itself rather than not having enough training examples. Still, uncertainty sampling did beat random selection, showing that picking the instances the model struggles with most is a better choice than choosing at random.

Question 4: Semi-Supervised Model Evaluation

4.1 Performance Comparison

Metric	Q1 Supervised (train-val split)	Q3 Semi-Supervised
Accuracy	0.8307	0.8298
<=50K Precision	0.8545	0.8542
<=50K Recall	0.9329	0.9318
<=50K F1	0.8920	0.8913
>50K Precision	0.7240	0.7204
>50K Recall	0.6254	0.5251
>50K F1	0.6089	0.6075

4.2 Changes in Model Confidence

The posterior ratio distribution remained almost identical after semi-supervised training. The mean log_R shifted marginally from 24.24 to 24.14 and the standard deviation moved less as well from 373.09 to 372.12. Out of 15087 test instances, only 38 (0.25%) actually changed their predicted class. The overall confidence of the model was unchanged by the addition of 200 labelled instances.

Model	Mean	Median	Std
Supervised	24.24	-5.63	373.09
Sem-Supervised	24.14	-5.65	372.12

4.3 Changes in Feature Parameters

The estimated means/variance have not changed significantly. Most shifted by less than 0.1, which is negligible. The largest change was capital-gain for >50K, where the mean went from 3542 to 3413, but with a standard deviation over 13000, that's less than a 1% shift. The class priors also stayed roughly the same, going from 75.4/24.6% to 74.8/25.2%. 200 new instances don't carry enough weight to move parameters that were already estimated from around 12000 training samples.

4.4 Changes in Most Predictive Features

The top predictive category values for each class remained identical after semi-supervised learning. Education continued to dominate the >50K predictions, with Prof-school, Doctorate and Masters holding the top three positions. For <=50K, relationship status (Own-child) and low-wage occupations (Priv-house-serv, Other-service) remained the strongest indicators. This was expected since the categorical probability ratios are computed from large counts in the training data and 200 additional instances do not shift those counts enough to reorder the rankings.

4.5 Discussion

The results were expected. Naive Bayes is a simple model with few parameters, so it doesn't need much data to learn what it can learn, however it still assumes features are unrelated, which is not true. For this dataset, a better model would likely help more than more labels and data.